Cambridge O Level Computer Science

Chapter 3: Hardware

Lecture notes with Check-Yourself questions and a full answer key.

In this chapter you will learn about:

- Computer architecture: the CPU, von Neumann architecture, registers, buses, the Fetch-Decode-Execute cycle, cores, cache and the clock, instruction sets, and embedded systems.
- Input and output devices: barcode and QR scanners, digital cameras, keyboards, microphones, mice, 2D and 3D scanners, touch screens, actuators, projectors, printers, screens, speakers, and sensors for control and monitoring.
- Data storage: primary storage (RAM and ROM), secondary storage (magnetic, optical and solid state), virtual memory and cloud storage.
- Network hardware: network interface cards, MAC addresses, IP addresses and routers.

3.1 Computer architecture

3.1.1 The central processing unit (CPU)

The central processing unit (CPU), also called the microprocessor or processor, is central to all modern computer systems, including tablets and smartphones. It is usually installed as an integrated circuit on a single microchip. The CPU executes and processes all the instructions and data in a computer application. It consists of:

- Control unit (CU): reads instructions from memory, decodes them, and sends control signals along the control bus telling every other component what to do. It keeps the flow of data and instructions synchronised throughout the computer.
- Arithmetic and logic unit (ALU): carries out arithmetic operations (addition, subtraction, logical shifts) and logic operations (AND, OR) while a program runs. Multiplication and division are done as sequences of additions/subtractions and shifts. A computer can have more than one ALU for specialist tasks.
- Registers: tiny, very fast storage locations inside the CPU that hold data and instructions temporarily during processing.
- Buses: parallel sets of wires carrying data, addresses and control signals.

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- System clock: produces the timing signals that keep everything synchronised; without it the computer would crash.

Check yourself

1. 1. State the three main components inside a CPU.
2. 2. Explain the difference between the jobs of the control unit and the arithmetic and logic unit.
3. 3. Why is the system clock essential to the operation of a computer?

3.1.2 Von Neumann architecture

In the mid-1940s John von Neumann proposed the stored program computer, the basis of computer architecture ever since. Its novel ideas were:

- Programs are stored in memory (not fed in while the machine runs).
- Stored programs are made of instructions executed in sequential order.
- The CPU can access memory directly.
- Memories store programs as well as data.

Registers

Registers can be general purpose or special purpose. The special purpose registers used in this book are:

Register	Abbreviation	Function
Program counter	PC	Holds the address of the next instruction to be fetched
Memory address register	MAR	Holds the address of the memory location being read from or written to
Memory data register	MDR	Holds data just read from memory, or data about to be written to memory
Current instruction register	CIR	Holds the current instruction while it is decoded and executed
Accumulator	ACC	Holds data temporarily during ALU calculations

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Reading from and writing to memory

Memory is divided into locations, each with a unique address and contents. The MAR and MDR work together for every access:

Worked example: Read the contents of memory location 1010, which holds 11001100.

The address 1010 is written into the MAR: MAR = 1010.

A read signal is sent to memory.

The contents of location 1010 are copied into the MDR: MDR = 11001100.

Worked example: Write the value 00110011 into memory location 1011.

The data is written into the MDR first: MDR = 00110011.

The address is written into the MAR: MAR = 1011.

A write signal is sent, and 00110011 is stored at location 1011.

Check yourself

4. 1. State the function of each of these registers: PC, MAR, MDR, CIR, ACC.
5. 2. A CPU wants to read memory location 1101, which contains 10101010. Describe the steps, naming the registers and signals used.
6. 3. Explain why the MAR is needed before any read or write operation.

System buses

Buses are parallel transmission components: each wire carries one bit. The three buses of the von Neumann architecture are:

- Address bus: carries addresses around the system. Between CPU and memory it is unidirectional (one direction only), so addresses cannot be carried back to the CPU.
- Data bus: bidirectional; carries data between CPU and memory and to and from input/output devices. The data can be an address, an instruction or a value.

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- Control bus: bidirectional; carries control signals from the control unit to all other components. It is usually 8 bits wide, which is enough since it only carries signals.

Bus width matters. The wider the address bus, the more memory locations can be addressed: a 16-bit address bus addresses 2^16 = 65,536 locations, while a 32-bit address bus addresses 2^32 = 4,294,967,296 locations. The wider the data bus, the larger the word length it can carry (a word is a group of bits treated as one unit, commonly 16, 32 or 64 bits), which improves performance.

Worked example: How many memory locations can a 20-bit address bus address?

Each of the 20 address lines can be 0 or 1, so the number of combinations is 2^20.

2^20 = 1,048,576 memory locations.

Check yourself

7. 1. Name the three system buses and state whether each is unidirectional or bidirectional.
8. 2. Explain why the address bus between CPU and memory is unidirectional.
9. 3. A computer has a 24-bit address bus. How many memory locations can it address?
10. 4. Explain how the width of the data bus affects computer performance.

The Fetch-Decode-Execute cycle

To run a program, the CPU fetches instructions and data from memory into registers, decodes each instruction, then executes it. The steps are:

11. The PC holds the address of the next instruction to be fetched.
12. This address is copied from the PC to the MAR using the address bus.
13. The instruction at that address is copied into the MDR.
14. The instruction is copied from the MDR into the CIR.
15. The PC is incremented by 1 so it points to the next instruction.
16. The instruction in the CIR is decoded and then executed: the control unit sends signals via the control bus to the appropriate components.
17. The cycle repeats while there are more instructions to service.

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